

Software Implementation Planning — roadmap

TOTAL: orig ~59d · forecast ~62d | PROGRESS: spent ~15.5d · rem ~46.5d | ~1d/week · +25% reserve

1

DONE

12 Sep 2026

Architecture baseline

TOTAL: orig ~1d · total ~0.7d

RESULT

- First software architecture baseline.
- TimingNode and application ownership are clear.
- Implementation can start deliberately.

DEMO

- Walk through the architecture diagrams.
- Explain TimingNode and application ownership.
- Show where the first executable fits.

DOCUMENT STATUS

10-SDP R-55

11-SIP R-60

30-UC O-25

31-SSSD W-45

60-SVP W-30

2

DONE

13 Sep 2026

Application core repository skeleton

TOTAL: orig ~6d · total ~1.8d

RESULT

- Application-core and runnable application artifacts.
- Clean Linux and Windows build/test.
- Traceable build identity and lifecycle.

DEMO

- Build from a clean checkout.
- Produce both artifacts.
- Start, identify and stop the application.

DOCUMENT STATUS

30-UC W-40

50-SDE-02 A-100

41-01-SSD R-25

43-01-SDD-02 W-55

32-03-IDD R-45

60-SVP W-35

3

DONE

29 Sep 2026

Application and API foundation

TOTAL: orig ~3d · total ~8.2d

RESULT

- SI-01 runs from external configuration.
- Public interfaces share application semantics.
- Black-box testing is repeatable.

DEMO

- Start SI-01 from configuration.
- Inspect version/status through public interfaces.
- Inspect IF-03 with the JavaFX test client.

DOCUMENT STATUS

30-UC W-55

41-01-SSD R-35

32-03-IDD R-65

32-11-IDD R-55

60-SVP W-45

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4

ACTIVE

end 12 Oct 2026

First registration-system slice

TOTAL: orig ~6d · forecast ~5.6d
PROGRESS: spent ~5.1d · rem ~0.5d

RESULT

- One TimingNode owns a durable, ordered registration stream.
- The same registration behaviour is available through the public API and Engineering Client.
- Restart rebuilds local registration history without inventing new records.

DEMO

- Configure a location, open the TimingNode and submit one accepted registration.
- Observe the committed record in history and as a live update.
- Close, restart and confirm that the committed history is recovered.

DOCUMENT STATUS

03-DOM W-80

30-UC R-70

41-01-SSD R-55

32-03-IDD R-70

60-SVP W-60

5

PLANNED

end 30 Nov 2026

Simulated antenna input and runtime behaviour

TOTAL: orig ~6d · forecast ~6d

RESULT

- Simulated antenna observations reach the normal registration path.
- Sustained input can be measured without bypassing TimingNode ownership.
- Restart/recovery works with the same simulated input path used by automated tests.

DEMO

- Open one TimingNode and feed repeatable tag observations through SimulatedAntenna.
- Show which observations become committed registrations and inspect the runtime counters.
- Restart SI-01 and continue using the same simulated input configuration.

DOCUMENT STATUS

30-UC R-80

41-01-SSD R-70

32-03-IDD R-80

60-SVP R-70

6

PLANNED

end 11 Jan 2027

Desktop GUI Application

TOTAL: orig ~4d · forecast ~5d

RESULT

- A separate operator GUI can connect to and operate SI-01 through the public API.
- Connection loss and stale state are visible instead of being hidden.
- SI-02 remains independently buildable from SI-01.

DEMO

- Connect SI-02 to a running SI-01 and operate one TimingNode.
- Show current status and registration history/live updates.
- Disconnect, reconnect and rebuild the current view.

DOCUMENT STATUS

30-UC R-85

41-02-SSD W-60

32-03-IDD R-85

60-SVP R-75

Software Implementation Planning — roadmap

TOTAL: orig ~59d · forecast ~62d | PROGRESS: spent ~15.5d · rem ~46.5d | ~1d/week · +25% reserve

7

PLANNED

end 05 Apr 2027

Backoffice and multi-node integration

TOTAL: orig ~7d · forecast ~9d

RESULT

- Multiple TimingNodes keep independent state and ordered data streams.
- Reference data can be received and used for local derived timing.
- Local registration continues through a backoffice outage and synchronisation can resume.

DEMO

- Start a small multi-node SI-01 test setup and load reference/start-time data.
- Feed simulated registrations and inspect node-specific derived timing and outbound data.
- Interrupt the backoffice service, continue local work, reconnect and reconcile.

DOCUMENT STATUS

30-UC R-90

41-01-SSD R-85

43-01-SDD-03 W-60

60-SVP R-85

8

PLANNED

end 10 May 2027

Target platform decision

TOTAL: orig ~4d · forecast ~4d

RESULT

- One target-platform direction is selected with its main risks understood.
- Required prototype hardware and any custom-board need are explicit.
- Step 9 can start without reopening the basic platform choice.

DEMO

- Compare the credible platform options against the known software/device needs.
- Show the selected hardware block diagram.
- Show the prototype parts/cost path and remaining platform risks.

DOCUMENT STATUS

10-SDP R-75

11-SIP R-80

50-SDE-01 W-60

60-SVP R-85

9

PLANNED

end 14 Jun 2027

Target bring-up and deployment proof

TOTAL: orig ~4d · forecast ~4d

RESULT

- SI-01 runs repeatably on the selected target.
- Target resource limits are based on measurements rather than desktop assumptions.
- The deployment/start/restart path is known before device integration begins.

DEMO

- Boot/provision the target and start SI-01.
- Connect SI-02 and run a representative simulated/reference-data workload.
- Show target measurements and a clean restart.

DOCUMENT STATUS

50-SDE-01 R-75

60-SVP R-88

70-01-SUM W-55

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10

PLANNED

end 06 Sep 2027

Real timing-device integration

TOTAL: orig ~10d · forecast ~10d

RESULT

- Real devices feed the same application/domain paths already proven with simulation.
- Device health and recovery are observable.
- Simulated tests remain usable as the fast regression baseline.

DEMO

- Run a representative real-device registration/input flow.
- Observe the resulting state/data through SI-02.
- Disconnect or reset one device and demonstrate the supported recovery behaviour.

DOCUMENT STATUS

30-UC R-95

41-01-SSD R-95

60-SVP R-95

70-01-SUM W-70

11

PLANNED

end 15 Nov 2027

Integrated system and field proof

TOTAL: orig ~8d · forecast ~8d

RESULT

- The representative integrated setup can run and recover from selected failures.
- Operators can observe the important system state through normal interfaces.
- Remaining hardening work is based on field/integration evidence.

DEMO

- Run a representative timing session from input through GUI/backoffice.
- Interrupt one external dependency or device.
- Recover and finish the session without losing committed timing history.

DOCUMENT STATUS

30-UC A-95

41-01-SSD A-95

60-SVP A-95

70-01-SUM A-90

11-SIP R-95